Reviewer #2, Concern #1

Reviewer comment:

1. Please formalize the "near-boundary" component of the operational review rule in Section 4 and Algorithm 1. The function g maps a letter rating to a decision class, such as IG/HY, and $\hat{r}$ is subsequently defined as the modal decision class. It is therefore unclear how the condition $\hat{r} \in B$ identifies a rating "near" the IG/HY boundary.

Author response:

- We agree. The original rule incorrectly defined boundary proximity using the modal IG/HY class, which cannot encode distance from the BBB−/BB+ boundary. We now define proximity using the modal letter rating and its notch distance from that boundary. Because instability also occurs away from the boundary, proximity is used as a priority indicator rather than a mandatory referral condition.

Author action:

- Algorithm 1 and Section 4 now define near-boundary status from the modal letter rating using the BBB−/BB+ midpoint, while referral is based separately on the dissent count $d \geq c_{\text{ref}}$.
- Section 4 removes the earlier claim that boundary proximity reproduces the WATCH stratum and removes the outdated "roughly halves to about 33%" statement.
- Section 4 now reports that requiring both disagreement and near-boundary status would cover 13.3% of profiles with recall 0.176; referring on any disagreement covers 88.9%.

Reviewer #2, Concern #2

Reviewer comment:

2. Please clarify the sampling convention underlying the reported matched-pair flip probabilities. The Methods and the captions of Figures 2–4 should state whether two draws may select the same elicitation cell and whether the pair may differ in specification, seed, or both. This clarification would make the 25.2% result and the self-consistency analysis fully reproducible.

Author response:

- We agree that the sampling convention required explicit definition. The reported 25.2% flip probability is the probability that two independent draws with replacement from the same profile's empirical elicitation distribution fall in different classes. The profile is fixed; the draws may differ in specification, seed, or both, and identical-cell draws count as agreement.
- Using sampling without replacement changes the pooled estimate only from 25.2% to 25.45%. We therefore retain the original measure but now label it explicitly as total elicitation instability, rather than attributing it solely to specification variation.

Author action:

- Section 6.3 and the Figure 2 caption now specify replacement, identical-cell treatment, permitted sources of variation, and the fixed-profile condition.
- Section 3.5 now labels the 25.2% estimate as total elicitation disagreement and separately reports 24.3% after collapsing seeds within specification and 6.9% for seed variation within an identical profile×specification cell.

- Figure 4 caption now states the actual binomial voting procedure used; the pair-sampling description was removed from Figure 3 because Fleiss κ does not use sampled pairs.

Reviewer #2, Concern #3

Reviewer comment:

3. Please harmonize the description of the analytical sample near the beginning of the paper. The abstract reports 45 credit-health profiles, 32 specifications, and three seeds, giving 4,320 credit-rating elicitations, whereas the Introduction describes a 90-item battery and 8,640 elicitations. Although Section 6.1 explains that the additional 45 directional items are not analyzed in this paper, this distinction should be stated at the first mention of the 8,640-cell experiment to avoid the impression of inconsistent sample sizes.

Author response:

- We agree that the full experimental battery and the analytical sample should be distinguished at first mention. The full battery contains two sets of 45 profiles, but this paper analyses only the 45 credit-health profiles: $45 \times 32 \times 3 = 4{,}320$ designed cells, of which 4,316 returned substantive outputs. The additional 45 directional profiles are archived but not analysed here.

Author action:

- Introduction, at the first description of the experiment, now distinguishes the 90-profile full battery from the 45-profile analytical sample.
- Methods now distinguish designed and realized sample sizes and note that four non-substantive cells occurred in the credit-health arm.

Reviewer #2, Concern #4

Reviewer comment:

4. In Table 4 and Section 3.6, the statement that approximately 33% agreement is "expected by chance" requires qualification. If this is a uniform three-class reference value, please label it as such. If it is intended as a chance-corrected baseline based on the observed class marginals, the corresponding calculation should be provided.

Author response:

- We agree. The earlier 33.3% value was only a uniform three-class reference and was not the correct chance benchmark for the observed marginals. Using the empirical machine and agency marginals, expected agreement is 48.9%, compared with observed agreement of 53.7%. The excess is therefore 4.9 percentage points (95% CI -7.9 to 17.1), with Cohen's $\kappa = 0.095$.
- Accordingly, this arm does not show agreement above the marginal-matching chance benchmark.

Author action:

- Section 3.6 and Table 4 now report observed agreement, marginal-matching expected agreement, excess agreement with its confidence interval, and Cohen's κ.
- Section 3.6 no longer interprets machine–agency agreement as evidence of rating accuracy.

Reviewer #3, Concern #1

Reviewer comment:

the authors have made efforts to address the previous review comments, including reframing the contribution as an evaluation-methodology case study, removing the economic translations, correcting the permutation test, adding a reasoning-architecture arm, and providing held-out-specification validation for the SIS… however, the revised manuscript still requires careful evaluation, and the following points must be considered before it can be further considered for publication. this manuscript is a research article, and it is evaluated as such.

the per-axis variance decomposition remains non-identified under the resolution-limited d-optimal fractional grid, yet the abstract, highlights, and conclusions still state that no controllable axis exceeds about 5% of rating variance. the authors now report aliasing diagnostics, including a maximum absolute correlation of 0.333 between a main effect and a two-factor interaction of other factors, and they correctly label the decomposition as descriptive. however, the headline claim of "no axis exceeds ~5%" is still presented as a substantive finding rather than as a model-dependent descriptive partition. because main effects can be partially aliased with omitted interactions, the per-axis point estimates cannot support even a descriptive ranking of axes without explicit bounds or a design that separates them. the added R^2 range of 0.083 to 0.119 across model specifications does not identify individual axes; it only shows that the elicitation axes jointly explain little variance under the assumed models. this remains important because the paper's central interpretation—that instability is spread across elicitation choices rather than concentrated in a tunable knob—depends in part on this decomposition.

required action: either remove all per-axis numeric claims from the abstract, highlights, and conclusions, or conduct a fully crossed or resolution-v sub-experiment on a reduced set of factors and items that can cleanly estimate main effects and relevant two-way interactions. if a new experiment is not feasible, provide formal bounds on each per-axis variance share under all aliasing structures consistent with the observed design matrix, or fit a bayesian hierarchical model with explicit shrinkage priors on interaction terms and report posterior intervals for the main effects. the abstract and highlights should then state only what is identified: that the elicitation axes jointly explain a small and model-dependent share of variance, not that any specific axis is bounded by ~5%.

Author response:

- We agree that the fractional design cannot identify individual elicitation-axis variance shares when omitted two-way interactions are allowed. We therefore removed the per-axis interpretation rather than attempting to bound or rank individual axes.
- Formally, omitted interactions imply

  $$\mathbb{E}[\hat{\beta}] = \beta + A\gamma,$$

  and the design places no finite restriction on $\|\gamma\|_2$; therefore no finite design-only bound exists for an individual main effect. Recomputed diagnostics on the actual 32-specification design confirm substantial aliasing: the maximum main-effect–interaction correlation is 0.504, the largest canonical correlation is 1.000, and six of eight exceed 0.9.
- The retained result is only that the elicitation axes jointly explain $R^2 = 0.083$–0.119 under the fitted models. The model-free disagreement results are unaffected.

Author action:

- Section 6.2 now states explicitly that individual elicitation-axis effects are not identified and reports the aliasing algebra, corrected design diagnostics, and canonical-correlation analysis.
- Abstract, Highlights, Results, and Discussion no longer state that no individual axis exceeds approximately

5%.

- Table 1 no longer reports individual-axis estimates as interpretable variance shares.

Reviewer #3, Concern #2

Reviewer comment:

the real-issuer arm still does not provide strong external validity, and the decontamination procedure remains insufficiently validated for the claims made... the perturbation applies a single per-issuer uniform scaling factor to all dollar quantities. this preserves financial ratios and Altman Z" by construction, but it does not prevent an LLM from inferring issuer identity from the pattern of ratios, nor does it rule out memorized issuer knowledge that is not based on exact figures. the fingerprinting test only measures exact-figure lookup. the authors acknowledge this limitation, but the abstract still states that specification variation is observed in "31 perturbed and fingerprint-screened real issuers," which can be read as external validation. in addition, the scale factor is confounded with issuer identity because each issuer receives only one draw. the sample of 31 issuers and 12 specifications is also small for any quantitative external claim.

required action: add a ratio-pattern identification test. for example, train a classifier on financial ratios to predict issuer identity from the perturbed profiles, or prompt the LLM to infer the issuer from the ratio profile after perturbation, and report the identification rate. conduct a within-issuer scale sensitivity design by re-eliciting a subset of issuers at multiple scale factors, such as 0.6, 1.0, and 1.7, so that scale is not fully confounded with issuer identity. if these checks show residual identification or if the sample remains too small, remove the real-issuer claim from the abstract and conclusion and present the arm only as an illustrative descriptive limitation. the paper should not imply external validity beyond what the design can support.

Author response:

- We agree that uniform scaling alone cannot establish decontamination because it preserves ratio patterns and originally confounded scale with issuer identity. We therefore performed both requested checks.
- First, all 31 issuers were re-elicited at scales 0.6, 1.0, and 1.7 using the frozen 12-specification grid. Extreme-scale classifications differed in 15.9% of cells versus 15.2% under the within-issuer permutation null ($p = 0.644$), providing no detectable incremental scale effect.
- Second, we tested issuer identification from ratio patterns. The open-ended probe produced no correct identifications, but the closed-set probe identified the correct issuer in 19.4% of substantive calls versus a 3.2% chance benchmark ($p = 5 \times 10^{-18}$). We therefore do not treat ratio-preserving scaling as decontamination and retain the real-issuer arm only as an illustrative descriptive check.

Author action:

- Section 3.7 and Table 5 now report the within-issuer scale experiment and ratio-pattern identification experiment.
- Abstract, Section 3.6, and Conclusion now describe the real-issuer arm as illustrative and descriptive and make no external-validity claim from it.
- Supplementary archive now includes the pre-registered experimental specification, frozen outputs, manifests, and analysis code.

Reviewer #3, Concern #3

Reviewer comment:

the SIS validation is improved, but its incremental value over simple disagreement-based baselines and its status as a deployable model-risk control remain unproven... at the binary IG/HY gate with a low threshold, the SIS flag reduces to "any disagreement" across the k elicitations. the reported Youden-optimal threshold of 0.08 is below the smallest non-zero SIS value at k=16, so the operating rule is essentially "flag any issuer with at least one disagreeing calibration elicitation." this rule is already directly observable from the k elicitations and does not require the continuous SIS. the manuscript does not compare SIS against simple baselines such as the number of dissenting elicitations, the modal rating's distance to the IG/HY boundary, self-consistency vote margins, semantic entropy, or token-level confidence. the threshold is also chosen and evaluated on the same held-out split, and cross-model transfer is moderate and inconsistent. this matters because the abstract and highlights present the SIS as a "validated" and "deployable model-risk control."

required action: benchmark SIS against these simple baselines in both held-out-specification and held-out-issuer settings. report a decision-curve analysis or net-benefit analysis for triage, showing whether SIS improves referral decisions over "flag any disagreement," over distance to the IG/HY boundary, and over self-consistency vote margins. if SIS does not provide incremental value beyond these baselines, reframe it as a descriptive dispersion index rather than a deployable control, and remove or substantially qualify the "validated" and "deployable" language in the abstract and highlights. if it does provide incremental value, report the threshold selection in a fully nested cross-validation or on an independent calibration set and show how performance varies with k and tau.

Author response:

- We agree that SIS must be compared directly with simpler disagreement measures. In both the held-out-specification analysis and the profile-split subsampling analysis, SIS provides no incremental ranking information over dissent count at the binary IG/HY gate. The previously reported Youden threshold $\tau = 0.08$ corresponds to $d \geq 3$ at the retrospective $k = 48$ and collapses to $d \geq 1$ at $k = 16$ (smallest non-zero index values 0.0408 and 0.117). It is no longer used, and the manuscript states the rule directly on d.
- At that resolution,
$$\text{SIS} = 2(m/k)(1 - m/k),$$
so SIS, dissent count, and vote margin are rank-equivalent. Empirically, each gives AUC 0.936, Spearman correlation 1.000, and identical net-benefit curves. At three bands the measures are not exactly rank-identical, but they produce the same AUC in the present analyses.
- We therefore no longer present SIS as a superior predictive instrument or use a tuned SIS threshold operationally. Referral is stated directly in terms of dissent count $d \geq c_{\text{ref}}$, while SIS is retained as a descriptive dispersion and governance representation. Boundary distance performs materially worse; discrete semantic entropy is again rank-equivalent at the binary gate.

Author action:

- Section 4.3 and Table 8 now benchmark SIS against dissent count, vote margin, boundary distance, and discrete semantic entropy in both the held-out-specification and profile-split analyses.
- Table 9 now reports net-benefit comparisons of the referral strategies against refer-all and refer-none.
- Sections 4.1–4.3 now present SIS as a descriptive dispersion index and governance representation, with the operational rule stated directly on dissent count.
- Abstract and Highlights no longer describe SIS as a validated or deployable model-risk control.

Reviewer #3, Concern #4

Reviewer comment:

under the resolution-limited d-optimal fractional grid, can you provide formal identifiable bounds on the maximum possible main effect of answer presentation and the provider-by-item interaction under all aliasing structures consistent with the observed design matrix, and how would those bounds change the abstract's claim that no controllable axis exceeds about 5% of rating variance?

Author response:

- The two quantities have different identification properties.
- For answer presentation, no finite design-only bound exists because omitted interaction effects enter the estimated main effect through

  $$\mathbb{E}[\hat{\beta}] = \beta + A\gamma,$$

  and the design provides no finite restriction on $\|\gamma\|_2$. The former approximately 4.6% estimate therefore cannot support an identified upper bound or ranking.
- The provider×item component is different because items are crossed with specifications. All 1,440 item×specification cells are represented; only four are missing one seed, and the largest realized correlation with a specification-axis contrast is 0.0022. We therefore retain the approximately 1.4% provider×item component, subject to this negligible imbalance.
- The manuscript now reports only the identified joint result: the elicitation axes explain $R^2 = 0.083$–0.119 under the fitted specifications.

Author action:

- Section 6.2 now provides the formal aliasing analysis and numerical diagnostics establishing why no finite design-only bound exists for individual elicitation-axis effects.
- Abstract and Highlights no longer contain the approximately 5% per-axis claim.
- Table 1 no longer reports individual-axis estimates as interpretable variance shares.